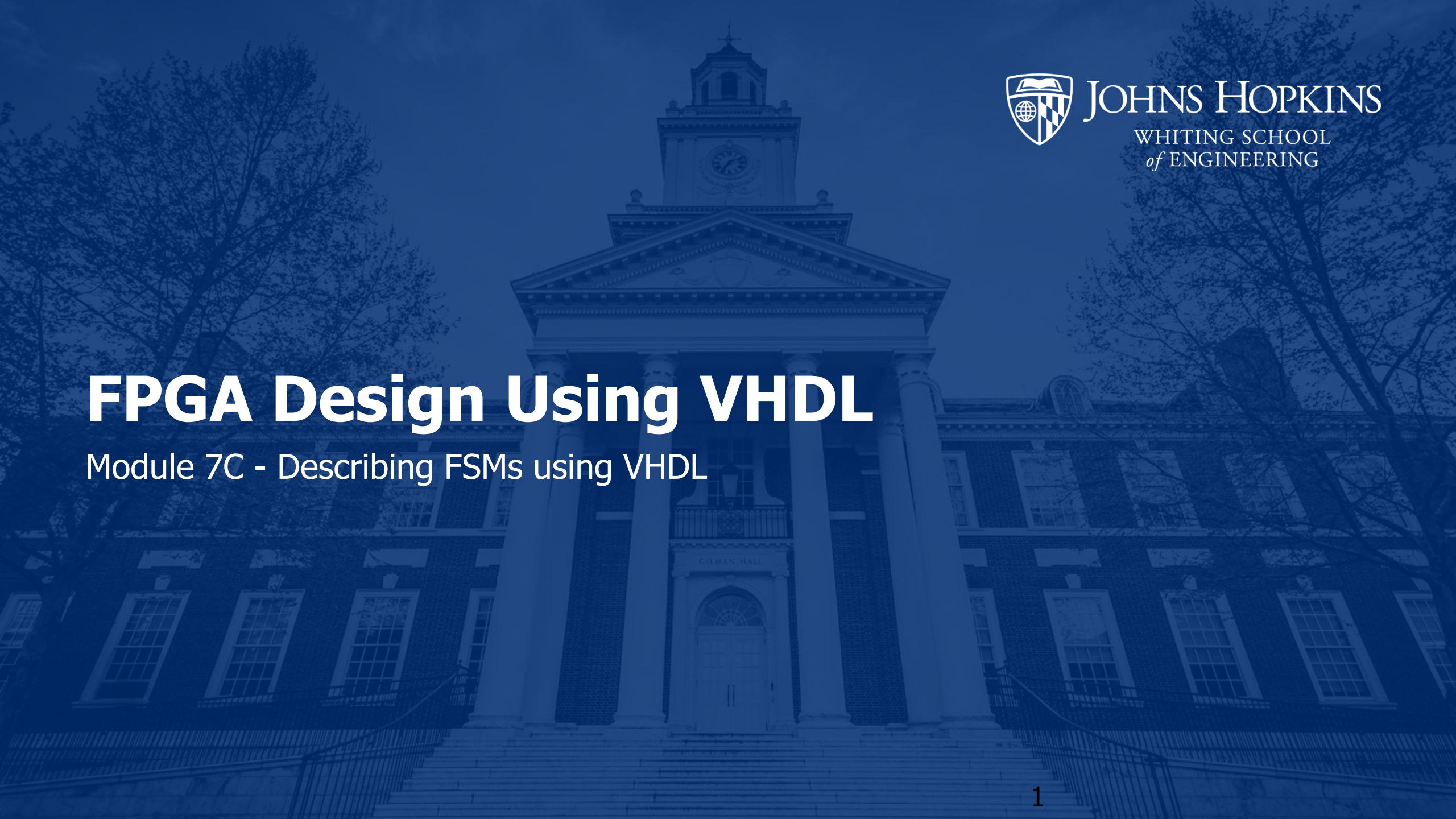


JOHNS HOPKINS
WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 7C - Describing FSMs using VHDL

FSMs in VHDL

- FSMs can be easily described with processes
- Synthesis tools understand FSM description if certain rules are followed
  - States are represented with enumerated types
  - State transitions should be described
    1. In a process sensitive to clock and asynchronous reset signals only

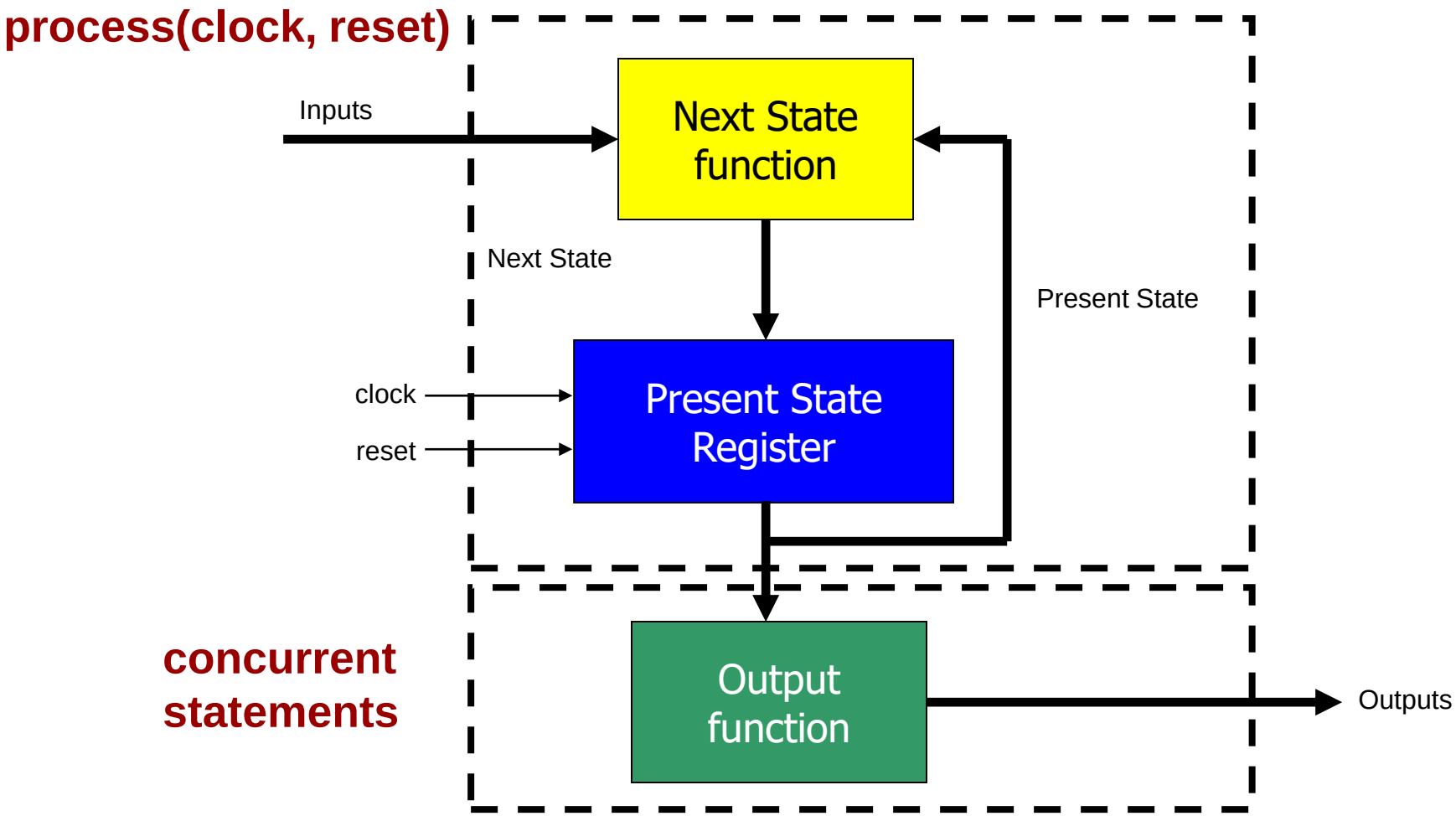
    Or alternatively

    2. In a process sensitive to current state and inputs only
  - Outputs can be described
    1. As concurrent statements outside the process

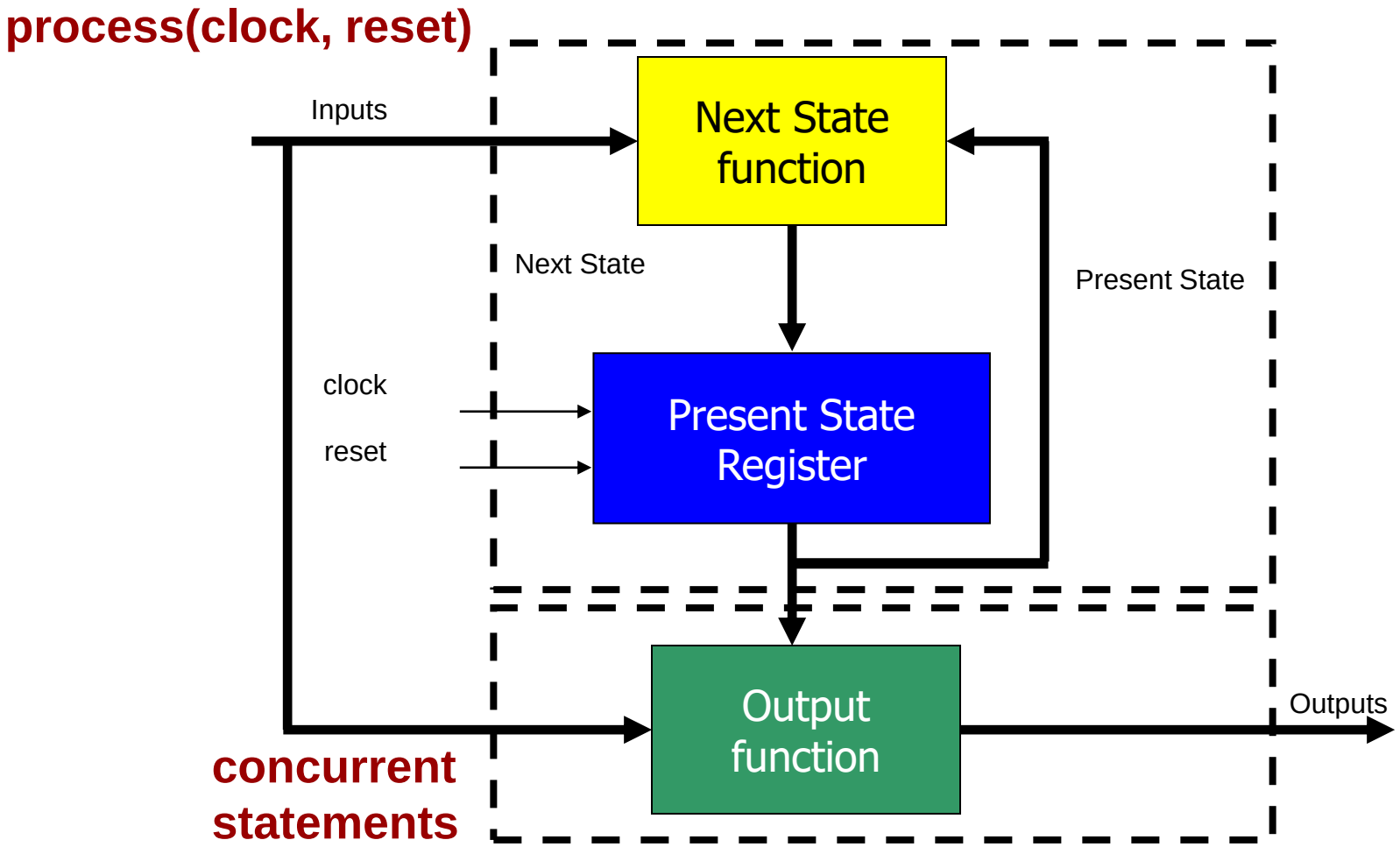
    Or alternatively

    2. In a process as part of the FSM encoding process

Moore FSM

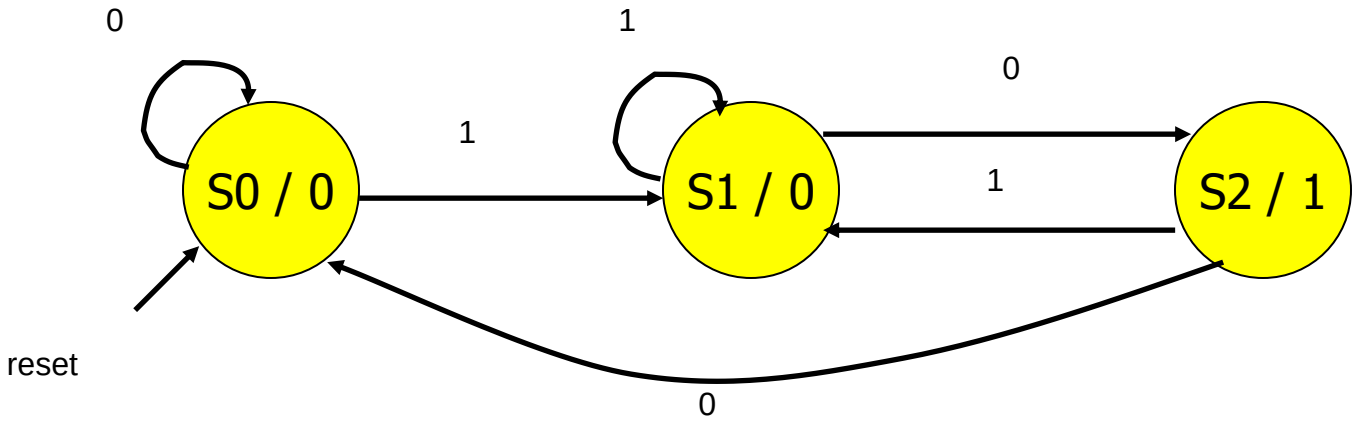

process(clock, reset)
Inputs
Next State function
Next State
Present State
clock
reset
Present State Register
concurrent statements
Output function
Outputs

Mealy FSM


process(clock, reset)
Inputs
Next State function
Next State
Present State
clock
reset
Present State Register
Output function
Outputs
concurrent statements

Moore FSM Example

- Moore FSM that Recognizes Sequence "10"

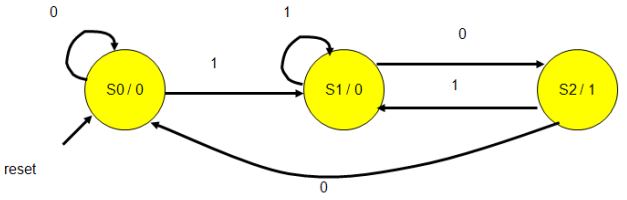


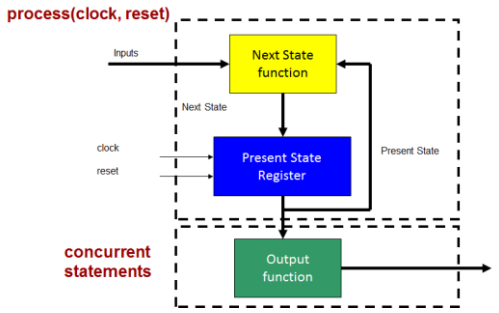
Moore FSM in VHDL

```
TYPE state_type IS (S0, S1, S2);
SIGNAL Moore_state: state_type;
⋮
U_Moore: PROCESS (clock, reset)
BEGIN
   IF(reset = '1') THEN
      Moore_state <= S0;
   ELSIF (rising_edge(clock)) THEN
      CASE Moore_state IS
      WHEN S0 =>
         IF FSM_in = '1' THEN
            Moore_state <= S1;
         ELSE
            Moore_state <= S0;
         END IF;
      WHEN S1 =>
         IF FSM_in = '0' THEN
            Moore_state <= S2;
         ELSE
            Moore_state <= S1;
         END IF;
```

```
 WHEN S2 =>
         IF FSM_in = '0' THEN
            Moore_state <= S0;
         ELSE
            Moore_state <= S1;
         END IF;
      END CASE;
   END IF;
END PROCESS;

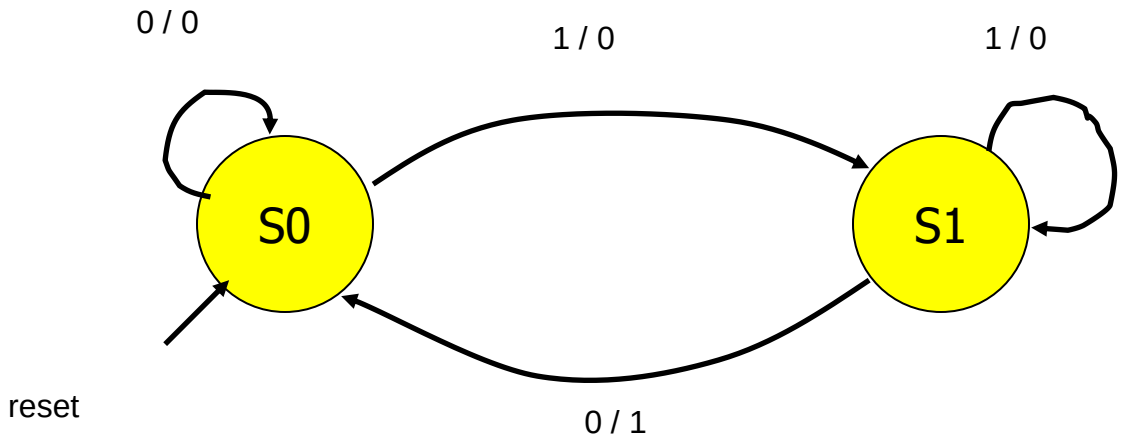
FSM_out <= '1' WHEN Moore_state = S2 ELSE '0';
```





Mealy FSM Example

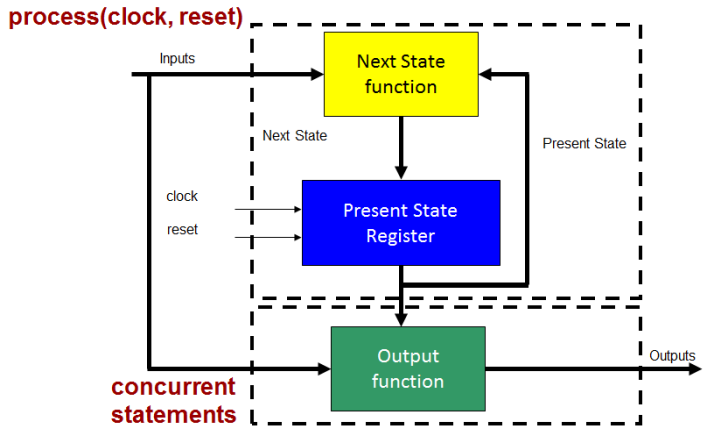
- Mealy FSM that Recognizes Sequence "10"

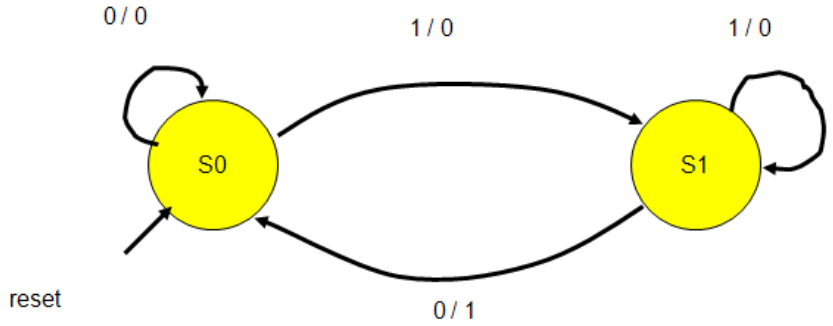


Mealy FSM in VHDL

```
TYPE state_type IS (S0, S1);
SIGNAL Mealy_state: state_type;
⋮
U_Mealy: PROCESS(clock, reset)
BEGIN
   IF(reset = '1') THEN
      Mealy_state <= S0;
   ELSIF (rising_edge(clock)) THEN
      CASE Mealy_state IS
      WHEN S0 =>
         IF FSM_in = '1' THEN
            Mealy_state <= S1;
         ELSE
            Mealy_state <= S0;
         END IF;
      WHEN S1 =>
         IF FSM_in = '0' THEN
            Mealy_state <= S0;
         ELSE
            Mealy_state <= S1;
         END IF;
      END CASE;
   END IF;
END PROCESS;

FSM_out <= '1' WHEN (Mealy_state = S1 AND FSM_in = '0') ELSE '0';
```





ModelSim Simulation of FSM

- Example of Moore and Mealy FSM to detect sequence "10"

